

Hrishi Kabra

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Education

Tulane University | School of Science and Engineering

August 2024 — May 2028

Bachelor of Science, Majors: Computer Science and Mathematics

New Orleans, LA

- Minors: Artificial Intelligence and Business | GPA: 3.95/4.0 | Dean's List | Alpha Kappa Psi Business Fraternity
- South Asian Scholar (merit-based full tuition award of ~\$67,000 per year), First Year Honors Scholar
- Relevant Coursework: Machine Learning, Multi Agent Systems, DSA, Linear Algebra, Computer Systems

Experience

Louisa AI

June 2026 — Present

Software Engineer Intern — Embedded AI Team

Mumbai, India

- Strengthened the relationship-intelligence graph by building a fallback pipeline (Python, OpenAI, Tavily, Firecrawl) that enriches no-LinkedIn contacts from open web sources to reconstruct role, career history, and education
- Designed a confidence-gated identity "judge" using LLM structured outputs that disambiguates same-name individuals, admitting only sources past a 0.80 match threshold to keep wrong-person data out of enriched records

Tulane Data and Artificial Intelligence

July 2025 — Present

Research Assistant

New Orleans, LA

- Co-authored paper accepted at AAMAS 2026 (Conference on Autonomous Agents and Multiagent Systems)
- Designed and optimized positional scoring models using simulated annealing, gradient descent, and mixed-integer programming (MIP) to optimize decision outcomes over large-scale preference datasets
- Designing a retrieval-stage method that reframes RAG top-k truncation as a proportional committee election, applying social-choice rules to improve evidence coverage and resolve conflicting sources in multi-agent RAG

Niyogin AI

June 2025 — July 2025

AI/ML Engineer Intern

Mumbai, India

- Developed and shipped a production-grade, end-to-end NLP pipeline using custom SpaCy named entity recognition (NER) to reliably extract structured information from noisy OCR output across financial and identity documents
- Achieved ~97% extraction precision and automated KYC and form digitization workflows to reduce manual review
- Engineered a Python-based document processing workflow to convert diverse multi-format and scanned bank statements into structured, query-ready Excel reports, streamlining downstream data ingestion and analysis

Tulane University Department of Computer Science

June 2025 — Present

Teaching Assistant — Intro to Data Science & Intro to Computer Science I

New Orleans, LA

- Led weekly labs and office hours for 50+ students, teaching Python, pandas, and core data analysis concepts

Projects

ReefScan | PyTorch, DINOv2, SAM2, Conformal Prediction, FastAPI, Next.js

December 2025 — Present

- Fine-tuned a DINOv2-B vision model in PyTorch to segment (SAM2) and classify coral colonies as healthy or bleached over 10,000+ images, reaching 89.5% test accuracy and 0.887 macro-F1 on a 1,565-image held-out split
- Beat zero-shot GPT-4o on the same test set by ~9 points accuracy and ~10 points macro-F1 with 3× better calibration (ECE 0.046 vs 0.152), showing a small specialist outperforming a frontier VLM on a narrow labeled task
- Wrapped predictions in split-conformal sets (90% coverage) to route uncertain colonies into an active-learning review queue; deployed with W&B training runs, FastAPI/Next.js, and a SQL-based observability dashboard

World Cup Prediction Market Engine | Python, SciPy, FastAPI, Next.js, TypeScript

June 2026 — Present

- Built a Dixon-Coles bivariate Poisson model fit via MLE on 15,494 international matches with a low-score correction, achieving a 0.572 Brier score versus a 0.613 Elo baseline on a walk-forward backtest against Euro 2024
- Implemented a path-independent Logarithmic Market Scoring Rule (LMSR) market maker pricing live odds over 100K Monte Carlo tournament simulations, deploying a FastAPI and Next.js stack on DigitalOcean and Vercel

Skills and Interests

Languages: Python, Java, C, C++, JavaScript, TypeScript, SQL, R, Assembly, HTML/CSS, Bash

Technologies: PyTorch, Pandas, NumPy, SciPy, FastAPI, Next.js, AWS (S3, Glue, Athena), Docker, SQLite, Git

Concepts: Statistical Modeling, Monte Carlo Simulation, Data Visualization, RAG, LLM Agents, NLP, Fine Tuning, MLOps, REST APIs, ETL & Data Pipelines, Data Structures & Algorithms, Full-Stack Development

Certifications: Anthropic Claude Code in Action, Anthropic Agent Skills & Subagents, OpenAI Agents & Workflows, LangChain Building Reliable Agents, PADI Master Scuba Diver, PADI Rescue Diver

Interests: Scuba Diving, Underwater Videography, Formula 1, Football, Catan, Monopoly